The asymmetric effects of exchange rate (USD/VND) on Vietnam's trade balance: Evidence from Nonlinear ARDL and Machine Learning models (2015-2025)

Bui Viet Huy
DSEB-MFE, NEU
Email: 11230542@st.neu.edu.com
Course: Time Series Analysis
Lecturer: Tran Thi Ha

Abstract—**This study investigates the asymmetric impact of the USD/VND real exchange rate (RER) on Vietnam's trade dynamics from January 2015 to December 2025. Utilizing a Nonlinear ARDL (NARDL) framework, we decompose RER movements to capture asymmetric transmission across three parallel models: aggregate trade balance, exports, and imports. The research advances existing literature by incorporating the post-COVID-19 recovery phase, oil price controls, and benchmarking NARDL's predictive performance against machine learning architectures (LSTM and Random Forest). Empirical findings confirm the J-curve effect and significant asymmetry, with VND depreciation exerting a more pronounced long-term impact on the trade balance than appreciation. The COVID-19 pandemic is identified as a significant structural disruptor, while robustness checks using REER validate the core results. Policy implications emphasize a 6–12 month adjustment lag for central bank interventions and the necessity for asymmetric risk-hedging strategies among export enterprises to mitigate non-linear currency shocks.**

Index Terms—**Exchange rate; Trade balance; NARDL; J-curve effect; Asymmetric effects; Vietnam; COVID-19, Machine learning**

I. Introduction

Vietnam's integration into the global economy represents a remarkable trajectory of export-led growth, with total trade turnover projected to reach approximately $405 billion by 2024, positioning the nation among the world's top 25 merchandise exporters. Central to this economic strategy is the managed floating exchange rate regime, where the USD/VND bilateral rate serves as the primary instrument of monetary transmission. The disproportionate significance of this specific currency pair stems from the fact that approximately 90% of Vietnam's international trade is invoiced and settled in US dollars. Consequently, the USD/VND real bilateral exchange rate (RER) functions as the definitive mechanism governing the international price competitiveness of Vietnamese commodities, effectively mediating the relationship between domestic policy and global trade outcomes.

However, the exchange rate–trade nexus has encountered unprecedented turbulence during the 2015–2025 period. The dual shocks of the COVID-19 pandemic and the subsequent aggressive tightening cycle by the US Federal Reserve exerted sustained depreciation pressure on the Vietnamese dong, culminating in historic lows and significant central bank interventions by early 2025. Despite the criticality of these dynamics, the existing empirical literature exhibits notable gaps. Prior research, including prominent studies by Truong and Vo (2023) and Nguyen et al. (2024), either fails to encompass the full post-pandemic recovery phase or omits vital control variables such as global oil prices and Foreign Direct Investment (FDI) within a unified framework. Furthermore, the absence of a simultaneous estimation across aggregate trade, export, and import dynamics prevents a holistic understanding of how asymmetric transmission mechanisms operate across different trade components.

To address these limitations, this study investigates three fundamental research questions: (1) Do the asymmetric effects of the RER on Vietnam's trade balance persist and intensify when the analysis accounts for recent structural shocks? (2) Is the J-curve hypothesis validated for Vietnam over the extended 2015–2025 sample period? and (3) Do machine learning architectures, specifically Long Short-Term Memory (LSTM) and Random Forest, achieve superior predictive accuracy compared to the traditional Nonlinear Autoregressive Distributed Lag (NARDL) framework? Utilizing a rigorous dataset of 132 monthly observations, the study employs the NARDL approach to decompose exchange rate movements into positive and negative partial sums, while systematically benchmarking econometric results against machine learning models to evaluate the trade-off between structural interpretability and predictive validity.

This research provides four distinct contributions to the empirical literature on exchange rate economics. First, it extends the temporal scope to the end of 2025, offering the first comprehensive analysis of Vietnam's trade dynamics that incorporates the post-COVID-19 stabilization era. Second, by introducing a structural dummy for the pandemic and integrating WTI crude oil prices as a control variable, the model effectively captures the impact of energy import dependence and cost-push inflation on trade flows. Third, the simultaneous estimation of three parallel NARDL models—aggregate trade balance, exports, and imports—enables a direct comparison of exchange rate asymmetry across all trade facets. Finally, by bridging the gap between econometric modeling and modern forecasting algorithms, this study provides policymakers with evidence-based insights into the relative merits of machine learning versus traditional structural models for trade balance management in an increasingly volatile global environment.

II. Literature Review

1. Theoretical Foundations of the Exchange Rate-Trade Balance Nexus

The classical starting point is the elasticity approach, especially the Marshall-Lerner condition. According to this condition, a currency depreciation improves the trade balance only when the combined price elasticities of export and import demand exceed one. This means that exchange rate policy is effective only if trade volumes respond sufficiently to changes in relative prices. In Vietnam, this mechanism is not automatic because exports are strongly linked to foreign-invested manufacturing, where imported intermediate inputs account for a large share of production. As a result, depreciation may raise the domestic-currency value of exports without generating a proportional increase in export volume. The J-curve hypothesis provides a dynamic explanation of why depreciation may first worsen the trade balance before improving it later. In the short run, existing contracts and slow quantity adjustment mean that import costs rise faster than export volumes respond. Over time, firms and consumers adjust to the new price structure, allowing exports to expand and imports to decline. This framework is particularly relevant for Vietnam because the economy is open, import-dependent and highly integrated into global value chains. Other theoretical perspectives also support the need for a broader exchange-rate framework. The purchasing power parity tradition treats the exchange rate as a long-run price-level adjustment mechanism, while the Balassa-Samuelson effect explains why fast-growing developing economies may experience real appreciation as productivity rises. The absorption approach adds a macroeconomic perspective by arguing that depreciation improves the trade balance only if national income rises relative to domestic absorption. Together, these theories suggest that the exchange rate-trade balance relationship depends not only on prices, but also on production structure, import dependence, domestic demand and long-run competitiveness.

2. Methodological Development: From ARDL to NARDL

The ARDL bounds testing approach developed by Pesaran, Shin and Smith (2001) is widely used in trade-balance studies because it can be applied when variables are a mixture of I(0) and I(1), performs well in small samples and estimates both short-run and long-run relationships within a single framework. These features make ARDL suitable for developing economies where macroeconomic time series are often limited and mixed in their integration properties. However, the standard linear ARDL model assumes symmetry: depreciation and appreciation are treated as equal in magnitude but opposite in direction. This assumption is restrictive because firms, consumers and policymakers may react differently to exchange rate increases and decreases. For example, exporters may not expand immediately after depreciation if they rely heavily on imported inputs, while importers may react more quickly to appreciation because foreign goods become cheaper. The NARDL framework proposed by Shin, Yu and Greenwood-Nimmo (2014) directly addresses this limitation by decomposing the exchange rate into positive and negative partial sums. This allows depreciation and appreciation to enter the model separately, so their short-run and long-run effects can differ. The framework also allows researchers to test asymmetry using Wald tests and to trace adjustment paths through dynamic multipliers. For studies of Vietnam's trade balance, NARDL is especially useful because it can test whether the J-curve is asymmetric rather than assuming a single uniform response to exchange rate changes.

3. Empirical Evidence on Exchange Rate Effects

International evidence generally supports the view that exchange rate effects are asymmetric. Bahmani-Oskooee and Fariditavana (2015) show that US bilateral trade balances respond differently to depreciation and appreciation, while Bahmani-Oskooee and Aftab (2017) find that asymmetric effects are clearer at the industry level than at the aggregate level. This suggests that aggregate models may hide sector-specific responses and underestimate the true importance of exchange rate movements. Evidence from ASEAN and emerging economies is also mixed but broadly consistent with the need for nonlinear modelling. Studies on Thailand, Malaysia, Indonesia and China show that the size, timing and direction of exchange-rate effects vary across countries, partners and industries. A recurring finding is that trade invoicing and vehicle-currency effects matter: when trade is invoiced mainly in USD, bilateral exchange rates alone may not fully capture the relevant price channel. This is important for Vietnam because the USD plays a dominant role in trade settlement and policy signalling.

4. Evidence from Vietnam

Vietnam provides a suitable context for studying asymmetric exchange-rate effects because its exchange rate regime is managed rather than freely floating, its export sector is strongly FDI-driven, and its trade structure is highly dependent on imported inputs. During 2015-2025, Vietnam experienced several exchange-rate and external shocks, including the 2015 RMB devaluation, US-China trade tensions, the COVID-19 period, the 2020 US currency-manipulator designation and the 2022-2024 Fed tightening cycle. These events make it unlikely that the exchange rate-trade balance relationship remained stable and symmetric throughout the whole period. Early studies on Vietnam mainly used VAR, IRF and linear ARDL models. Hoang (2013), Phan and Jeong (2015), Thom (2017) and Nguyen et al. (2017) generally find evidence of a J-curve pattern: depreciation tends to worsen the trade balance in the short run before improvement appears later. These findings provide an important foundation, but they are limited because linear models impose symmetry between depreciation and appreciation. More recent NARDL studies provide stronger evidence of asymmetry. Nguyen, Tran and Le (2023) find that VND depreciation worsens the trade balance in the short run but improves it in the long run, while appreciation effects are weaker or statistically insignificant. Thanh and Dinh (2021) show that including the USD as a vehicle currency increases the number of significant exchange-rate effects in Vietnam-EU trade, confirming the importance of USD-based channels. Nguyen and Huan (2018) also find that industry-level NARDL models reveal asymmetric responses that are not visible in aggregate ARDL models. Overall, the Vietnam literature supports the use of NARDL and highlights the risks of relying only on aggregate linear specifications.

5. Machine Learning as a Complementary Forecasting Approach

Although NARDL is valuable for identifying asymmetric long-run and short-run relationships, it remains limited by its pre-specified functional form and relatively small set of regressors. It is less flexible in capturing complex interactions among exchange rates, industrial production, FDI, external demand and global price conditions. These limitations are important for the 2015-2025 period, when Vietnam's trade balance was shaped by several structural shocks and nonlinear macroeconomic dynamics. Machine learning methods can complement NARDL by improving predictive performance and detecting nonlinear relationships that traditional econometric models may miss. Linear Regression can serve as a transparent benchmark model, while Random Forest and XGBoost are more flexible ensemble methods capable of capturing nonlinear effects and interactions among variables. Recent macroeconomic forecasting studies show that these models often outperform traditional linear time-series models, especially under uncertainty and structural instability. The main concern with machine learning is interpretability. This can be addressed through explainable AI tools such as SHAP values, which decompose each prediction into feature-level contributions. In this study, SHAP can help evaluate whether exchange rate depreciation, appreciation, REER, industrial production, FDI and other macroeconomic variables are truly important predictors of Vietnam's trade balance. Therefore, the combined NARDL-machine learning approach allows the study to balance structural explanation with predictive flexibility.

6. Research Gap and Contribution

The literature reveals four main gaps. First, most NARDL studies on Vietnam use data ending before 2021, meaning they do not fully capture the COVID-19 shock, the currency-manipulator episode and the Fed tightening cycle. Extending the sample to 2015-2025 allows the present study to analyse a more recent and structurally different period. Second, previous Vietnam studies have not systematically integrated NARDL with machine learning. This study contributes by combining an econometric model that identifies asymmetric exchange-rate effects with forecasting models that capture nonlinear interactions. Third, the role of REER remains underexplored compared with the bilateral USD/VND rate, even though Vietnam trades extensively with China, Korea, the EU and other non-USD partners. Including both USD/VND and REER therefore strengthens the analysis of external competitiveness. Fourth, earlier studies rarely examine whether asymmetry changes across sub-periods. This study addresses that gap by considering structural breaks and rolling-window dynamics.

III. Data and Methodology

1. Data Sources and Variable Description

This study employs monthly time-series data for Vietnam over the period from January 2015 to December 2025, yielding a total of 132 observations. The monthly frequency is selected because the exchange rate–trade balance relationship is expected to involve short-run adjustment lags, particularly under the J-curve mechanism. Compared with quarterly or annual data, monthly observations allow the model to capture adjustment dynamics within the 3–12 month window, during which import contract rigidity, delayed export response, and gradual quantity adjustment may occur.

The dependent variable in the baseline model is the trade balance, measured as the ratio of exports to imports. A trade balance ratio greater than one indicates that exports exceed imports, while a value below one indicates a trade deficit. In logarithmic terms, a positive value of $\ln TB_t$ reflects a trade surplus, whereas a negative value suggests that imports exceed exports. In addition to the aggregate trade balance model, this study estimates separate export and import models to identify whether exchange rate movements affect the trade balance through the export channel, the import channel, or both.

The key explanatory variable is the real bilateral exchange rate between the Vietnamese dong and the US dollar. The real bilateral exchange rate is computed as follows:

$$RER_t = NER_t \left(\frac{CPI_{US,t}}{CPI_{VN,t}} \right). \tag{1}$$

where NER_t denotes the monthly average nominal VND/USD exchange rate, $CPI_{US,t}$ is the US consumer price index, and $CPI_{VN,t}$ is Vietnam's consumer price index. Both CPI series are rebased to 2015 = 100 before constructing the real exchange rate. Under this definition, an increase in RER_t indicates a real depreciation of the Vietnamese dong, while a decrease indicates a real appreciation.

The empirical model also includes several control variables. Foreign direct investment is included because Vietnam's export sector is strongly linked to foreign-invested enterprises. Vietnam's industrial production index captures domestic production capacity and import demand. World industrial production growth is included as a proxy for global demand conditions. The change in domestic money supply controls for monetary conditions, while the WTI crude oil price captures global commodity price shocks. Finally, a COVID-19 dummy variable is included to account for the disruption caused by the pandemic period.

2. Variable Description

To ensure readability in the two-column format, the variable description is divided into two compact tables. Table I presents

TABLE I
MAIN VARIABLES

Variable	Definition	Sign
$\ln TB_t$	Logarithm of trade balance ratio, measured as exports divided by imports.	Dep.
$\ln EX_t$	Logarithm of total exports.	Dep.
$\ln IM_t$	Logarithm of total imports.	Dep.
ln_RER_pos$_t$	Positive partial sum of real exchange rate changes, capturing real depreciation.	$+$ for TB, EX
ln_RER_neg$_t$	Negative partial sum of real exchange rate changes, capturing real appreciation.	Asym.

TABLE II
CONTROL VARIABLES

Variable	Definition	Sign
$\ln FDI_t$	Logarithm of foreign direct investment.	$+$
IPI_VN$_t$	Vietnam's industrial production index.	$+$
IPI_World_GR_diff$_t$	Change in world industrial production growth.	$+$
ln_M2_diff$_t$	Change in domestic money supply.	Amb.
ln_WTI$_t$	Logarithm of WTI crude oil price.	Amb.
COVID$_t$	Dummy variable for the COVID-19 shock period.	$-$

the dependent variables and exchange rate variables, while Table II reports the control variables.

3. Descriptive Statistics

Table III and Table IV report the descriptive statistics of all variables. The first table presents the mean, standard deviation, minimum, and maximum values, while the second table reports skewness and kurtosis. Splitting the descriptive statistics into two tables prevents overflow in the two-column format and improves readability.

These statistics provide an initial overview of the distributional characteristics of the dataset before formal econometric estimation. For example, if the mean value of the trade balance ratio is below one, or if the mean of its logarithmic transformation is negative, this suggests that Vietnam recorded trade deficits on average during part of the sample period. High dispersion in the exchange rate or trade variables may indicate substantial macroeconomic volatility. Skewness and kurtosis are also useful for identifying whether the variables are symmetrically distributed or affected by extreme observations, particularly during periods such as the COVID-19 shock and the global monetary tightening cycle.

4. Theoretical Model: Two-Country Trade Framework

Following the standard two-country trade model, the trade balance is theoretically determined by the real exchange rate, domestic economic activity, and foreign demand. In this framework, domestic and foreign goods are imperfect substitutes, and trade flows respond to relative prices and income levels. The reduced-form trade balance equation can be written as:

$$\ln TB_t = f(\ln RER_t, \ln Y_{VN,t}, \ln Y_{W,t}). \quad (2)$$

TABLE III
DESCRIPTIVE STATISTICS I

Variable	Mean	Std.	Min	Max
$\ln TB_t$				
$\ln EX_t$				
$\ln IM_t$				
ln_RER_pos$_t$				
ln_RER_neg$_t$				
$\ln FDI_t$				
IPI_VN$_t$				
IPI_World_GR_diff$_t$				
ln_M2_diff$_t$				
ln_WTI$_t$				
COVID$_t$				

TABLE IV
DESCRIPTIVE STATISTICS II

Variable	Skew.	Kurt.
$\ln TB_t$		
$\ln EX_t$		
$\ln IM_t$		
ln_RER_pos$_t$		
ln_RER_neg$_t$		
$\ln FDI_t$		
IPI_VN$_t$		
IPI_World_GR_diff$_t$		
ln_M2_diff$_t$		
ln_WTI$_t$		
COVID$_t$		

where TB_t is the trade balance ratio, RER_t is the real exchange rate, $Y_{VN,t}$ represents domestic income, and $Y_{W,t}$ represents foreign income or world demand. In this framework, the real exchange rate captures the relative price effect, domestic income captures import demand pressure, and foreign income captures external demand for Vietnamese exports.

A rise in the real exchange rate represents real depreciation of the Vietnamese dong. Theoretically, depreciation may improve the trade balance by making exports cheaper and imports more expensive. However, this effect may not be immediate because exports and imports may respond differently to favourable and unfavourable exchange rate movements. Therefore, imposing symmetry may produce biased or incomplete estimates of the exchange rate–trade balance relationship.

5. Nonlinear ARDL Model and Partial Sum Decomposition

To address the limitation of symmetry in the linear ARDL framework, this study applies the nonlinear autoregressive distributed lag model developed by Shin et al. (2014). The NARDL model allows the real exchange rate to be decomposed into positive and negative partial sums, thereby capturing asymmetric long-run and short-run effects. The positive partial sum represents real depreciation of the Vietnamese dong, while the negative partial sum represents real appreciation.

Following Shin et al. (2014), changes in the real exchange rate are decomposed as follows:

$$\text{ln\_RER\_pos}_t = \sum_{j=1}^{t} \max(\Delta \ln RER_j, 0). \quad (3)$$

$$\text{ln\_RER\_neg}_t = \sum_{j=1}^{t} \min(\Delta \ln RER_j, 0). \quad (4)$$

This decomposition allows real depreciation and real appreciation to have different long-run and short-run effects on the trade balance, exports, and imports.

6. Model Specification

To keep the empirical equations compact within the two-column format, the control variables are grouped into the following vector:

$$\begin{aligned} Z_t = [\ln FDI_t,\ \text{IPI\_VN}_t,\ \text{IPI\_World\_GR\_diff}_t, \\ \text{ln\_M2\_diff}_t,\ \text{ln\_WTI}_t,\ \text{COVID}_t]'. \end{aligned} \quad (5)$$

The first empirical model uses the trade balance as the dependent variable:

$$\begin{aligned} \ln TB_t = \alpha_0 + \alpha_1 \text{ln\_RER\_pos}_t + \alpha_2 \text{ln\_RER\_neg}_t \\ + \boldsymbol{\delta}' Z_t + \varepsilon_t. \end{aligned} \quad (6)$$

The second empirical model uses exports as the dependent variable:

$$\begin{aligned} \ln EX_t = \beta_0 + \beta_1 \text{ln\_RER\_pos}_t + \beta_2 \text{ln\_RER\_neg}_t \\ + \boldsymbol{\phi}' Z_t + u_t. \end{aligned} \quad (7)$$

The third empirical model uses imports as the dependent variable:

$$\begin{aligned} \ln IM_t = \gamma_0 + \gamma_1 \text{ln\_RER\_pos}_t + \gamma_2 \text{ln\_RER\_neg}_t \\ + \boldsymbol{\psi}' Z_t + v_t. \end{aligned} \quad (8)$$

In these models, ln_RER_pos_t measures the effect of real depreciation, while ln_RER_neg_t measures the effect of real appreciation. The use of separate positive and negative exchange rate components allows the study to examine whether depreciation and appreciation have asymmetric effects on Vietnam's trade balance, exports, and imports.

7. Expanded Model Specification

For clarity, the three models can also be written in expanded form. The trade balance model is specified as follows:

$$\begin{gathered} \ln TB_t = \alpha_0 + \alpha_1 \text{ln\_RER\_pos}_t \\ + \alpha_2 \text{ln\_RER\_neg}_t + \alpha_3 \ln FDI_t + \alpha_4 \text{IPI\_VN}_t \\ + \alpha_5 \text{IPI\_World\_GR\_diff}_t + \alpha_6 \text{ln\_M2\_diff}_t + \alpha_7 \text{ln\_WTI}_t \\ + \alpha_8 \text{COVID}_t + \varepsilon_t. \end{gathered} \quad (9)$$

The export model is specified as follows:

$$\begin{aligned} \ln EX_t = \beta_0 &+ \beta_1 \text{ln\_RER\_pos}_t + \beta_2 \text{ln\_RER\_neg}_t \\ &+ \beta_3 \ln FDI_t + \beta_4 \text{IPI\_VN}_t + \beta_5 \text{IPI\_World\_GR\_diff}_t \\ &+ \beta_6 \text{ln\_M2\_diff}_t + \beta_7 \text{ln\_WTI}_t + \beta_8 \text{COVID}_t + u_t. \end{aligned} \quad (10)$$

The import model is specified as follows:

$$\begin{aligned} \ln IM_t = \gamma_0 &+ \gamma_1 \text{ln\_RER\_pos}_t + \gamma_2 \text{ln\_RER\_neg}_t \\ &+ \gamma_3 \ln FDI_t + \gamma_4 \text{IPI\_VN}_t + \gamma_5 \text{IPI\_World\_GR\_diff}_t \\ &+ \gamma_6 \text{ln\_M2\_diff}_t + \gamma_7 \text{ln\_WTI}_t + \gamma_8 \text{COVID}_t + v_t. \end{aligned} \quad (11)$$

8. General NARDL Error Correction Specification

For each dependent variable $Y_t \in \{\ln TB_t, \ln EX_t, \ln IM_t\}$, the general NARDL error correction specification is written as:

$$\begin{aligned} \Delta Y_t = c &+ \rho Y_{t-1} + \theta_1 \text{ln\_RER\_pos}_{t-1} + \theta_2 \text{ln\_RER\_neg}_{t-1} \\ &+ \boldsymbol{\theta}_3' Z_{t-1} + \sum_{i=1}^{p} \lambda_i \Delta Y_{t-i} \\ &+ \sum_{i=0}^{q_1} \pi_i^{pos} \Delta \text{ln\_RER\_pos}_{t-i} \\ &+ \sum_{i=0}^{q_2} \pi_i^{neg} \Delta \text{ln\_RER\_neg}_{t-i} \\ &+ \sum_{i=0}^{q_3} \boldsymbol{\omega}_i' \Delta Z_{t-i} + e_t. \end{aligned} \quad (12)$$

where Y_t represents the dependent variable in each of the three models, and Z_t denotes the vector of control variables. The parameters θ_1 and θ_2 capture the long-run asymmetric effects of real depreciation and real appreciation, while π_i^{pos} and π_i^{neg} capture their short-run asymmetric effects.

9. Estimation Procedure

The estimation procedure follows an extended Box-Jenkins logic, beginning with the examination of time-series properties, followed by lag selection, cointegration testing, nonlinear ARDL estimation, and diagnostic checking. This sequential procedure ensures that the empirical model is statistically valid and that the estimated coefficients can be interpreted reliably.

10. Step 1: Unit Root Tests

The first step is to conduct unit root tests for all variables. This study applies both the Augmented Dickey-Fuller test and the Phillips-Perron test to examine whether each variable is stationary at level, $I(0)$, or becomes stationary after first differencing, $I(1)$. This step is essential because the ARDL and NARDL frameworks are valid only when the variables are integrated of order zero or one. If any variable is integrated of order two, $I(2)$, the bounds testing procedure is no longer appropriate.

The null hypothesis of both the ADF and PP tests is that the series contains a unit root. Rejection of the null hypothesis indicates stationarity.

TABLE V
UNIT ROOT TESTS AT LEVEL

Variable	ADF	PP
$\ln TB_t$		
$\ln EX_t$		
$\ln IM_t$		
ln_RER_pos_t		
ln_RER_neg_t		
$\ln FDI_t$		
IPI_VN_t		
$\text{IPI\_World\_GR\_diff}_t$		
ln_M2_diff_t		
ln_WTI_t		
COVID_t		

TABLE VI
UNIT ROOT TESTS AT FIRST DIFFERENCE

Variable	ADF	PP
$\Delta \ln TB_t$		
$\Delta \ln EX_t$		
$\Delta \ln IM_t$		
$\Delta\text{ln\_RER\_pos}_t$		
$\Delta\text{ln\_RER\_neg}_t$		
$\Delta \ln FDI_t$		
$\Delta\text{IPI\_VN}_t$		
$\Delta\text{IPI\_World\_GR\_diff}_t$		
$\Delta\text{ln\_M2\_diff}_t$		
$\Delta\text{ln\_WTI}_t$		
ΔCOVID_t		

11. Step 2: Optimal Lag Selection

The second step is optimal lag selection. Since the study uses monthly data with 132 observations, the maximum lag length is set at four months to preserve degrees of freedom and avoid overfitting. The optimal lag structure is selected using the Akaike Information Criterion. The Auto-ARDL procedure is employed to identify the most appropriate lag combination for the dependent variable and each explanatory variable.

12. Step 3: Bounds Test for Cointegration

The third step is the bounds test for cointegration. The ARDL bounds testing approach developed by Pesaran et al. (2001) is suitable when the regressors are a mixture of $I(0)$ and $I(1)$, provided that none of them is $I(2)$. The null hypothesis is that there is no long-run relationship:

$$H_0 : \rho = \theta_1 = \theta_2 = 0. \tag{13}$$

In the full model, the joint restriction also includes the lagged control variables contained in Z_{t-1}. If the calculated F-statistic exceeds the upper critical bound, the null hypothesis is rejected, indicating the existence of cointegration.

13. Step 4: NARDL Estimation

Once cointegration is confirmed, the nonlinear ARDL model is estimated to obtain both long-run and short-run coefficients. The long-run coefficients show the equilibrium effects of real depreciation and real appreciation on the trade balance, exports, and imports. The short-run coefficients capture immediate and lagged responses to changes in the real exchange rate.

The error correction term, ECM_{t-1}, is expected to be negative and statistically significant. A negative ECM coefficient confirms that deviations from the long-run equilibrium are corrected over time. The absolute value of the ECM coefficient indicates the speed of adjustment.

14. Step 5: Wald Test for Asymmetry

Asymmetry is tested using the Wald test. The null hypothesis of long-run symmetry is:

$$H_0 : \theta_1 = \theta_2. \tag{14}$$

Short-run asymmetry is tested by comparing the cumulative short-run coefficients of the positive and negative exchange rate changes:

$$H_0 : \sum_i \pi_i^{pos} = \sum_i \pi_i^{neg}. \tag{15}$$

If the null hypothesis is rejected, the results support the existence of asymmetric exchange rate effects.

15. Step 6: Diagnostic Tests

The final step is diagnostic checking. Several post-estimation tests are conducted to ensure the reliability of the estimated model. The Breusch-Godfrey LM test is used to detect serial correlation in the residuals. The ARCH test examines whether the residuals suffer from heteroscedasticity. The Ramsey RESET test assesses whether the model is correctly specified. Finally, the CUSUM and CUSUMSQ tests are used to evaluate parameter stability over the sample period.

These stability tests are particularly important because the sample includes major structural shocks, including the COVID-19 period and later external trade and monetary shocks. A model passes the diagnostic stage if there is no serial correlation, no ARCH effect, no evidence of functional misspecification, and the cumulative sum statistics remain within the critical bounds.